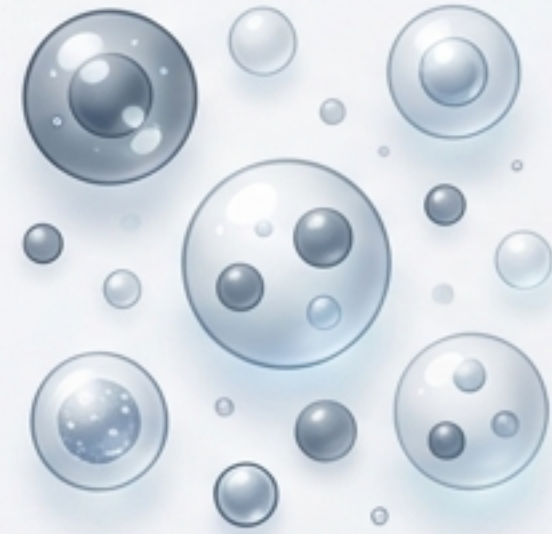


The Blueprint of Matter

Unlocking the rules of water, heat, and phase changes.



Heat Gain Processes

Heat Loss Processes

What defines a gas?

- ✓ It does not have a definite shape.
- ✓ It occupies space.
- ✓ It takes the shape of the container it is in.
- ✓ It can be compressed to occupy a smaller volume.

The Matrix of Matter



	Solid (Ice)	Liquid (Water)	Gas (Steam/Vapour)
Shape	Definite	Indefinite (takes shape of container)	Indefinite
Volume	Definite	Definite	Indefinite
Compressibility	Cannot be compressed	Cannot be compressed	Can be compressed

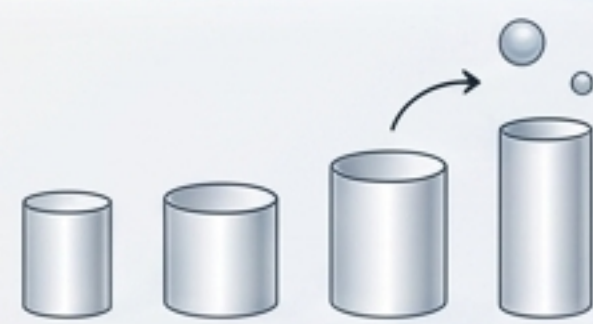
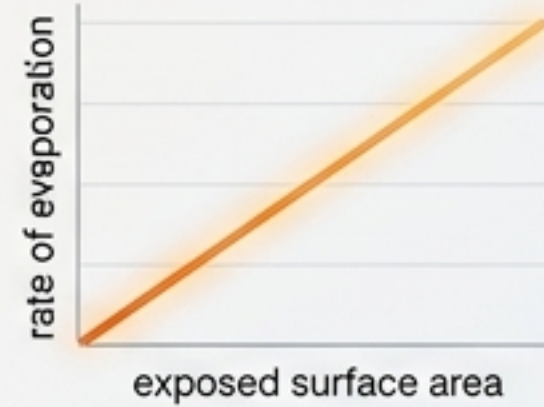
The Evaporation Control Panel

Evaporation (Liquid to Gas) happens at any temperature, but its rate can be controlled.

Exposed Surface Area



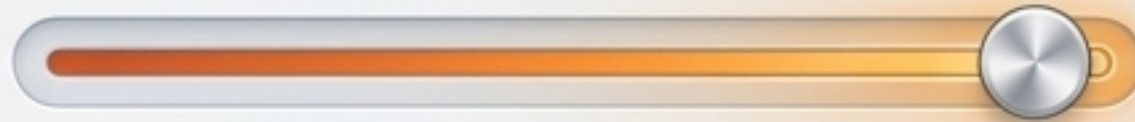
HIGH



Widest opening = fastest evaporation

Temperature

LOW



HIGH

Wind

LOW



HIGH

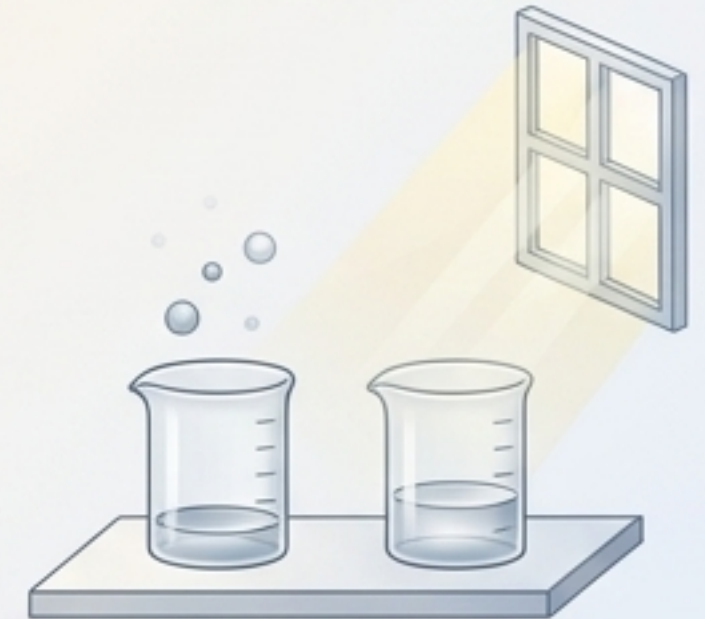
Humidity

LOW



LOW

Case Study



Evaporation occurs passively in both, regardless of the liquid type.

Case Study: The Cup Challenge

100ml of water placed in three identical cups under different conditions for 5 hours.

1	Cup X			Sunny (High Temp) + Windy	Maximum Evaporation
2	Cup Y			Sunny (High Temp) + Not Windy	Moderate Evaporation
3	Cup Z			Cloudy (Lower Temp) + Windy	Moderate Evaporation

The Aha Moment

Because evaporation happens to all liquids exposed to air, the baseline truth is revealed. The volume of water in each cup is definitively less than 100ml.

Two Paths to a Gas

Heat Gain

Evaporation



- **Change:** Liquid to Gas.
- **Location:** Occurs only at the exposed surface.
- **Temperature:** Occurs at any temperature.

Heat Gain

Boiling



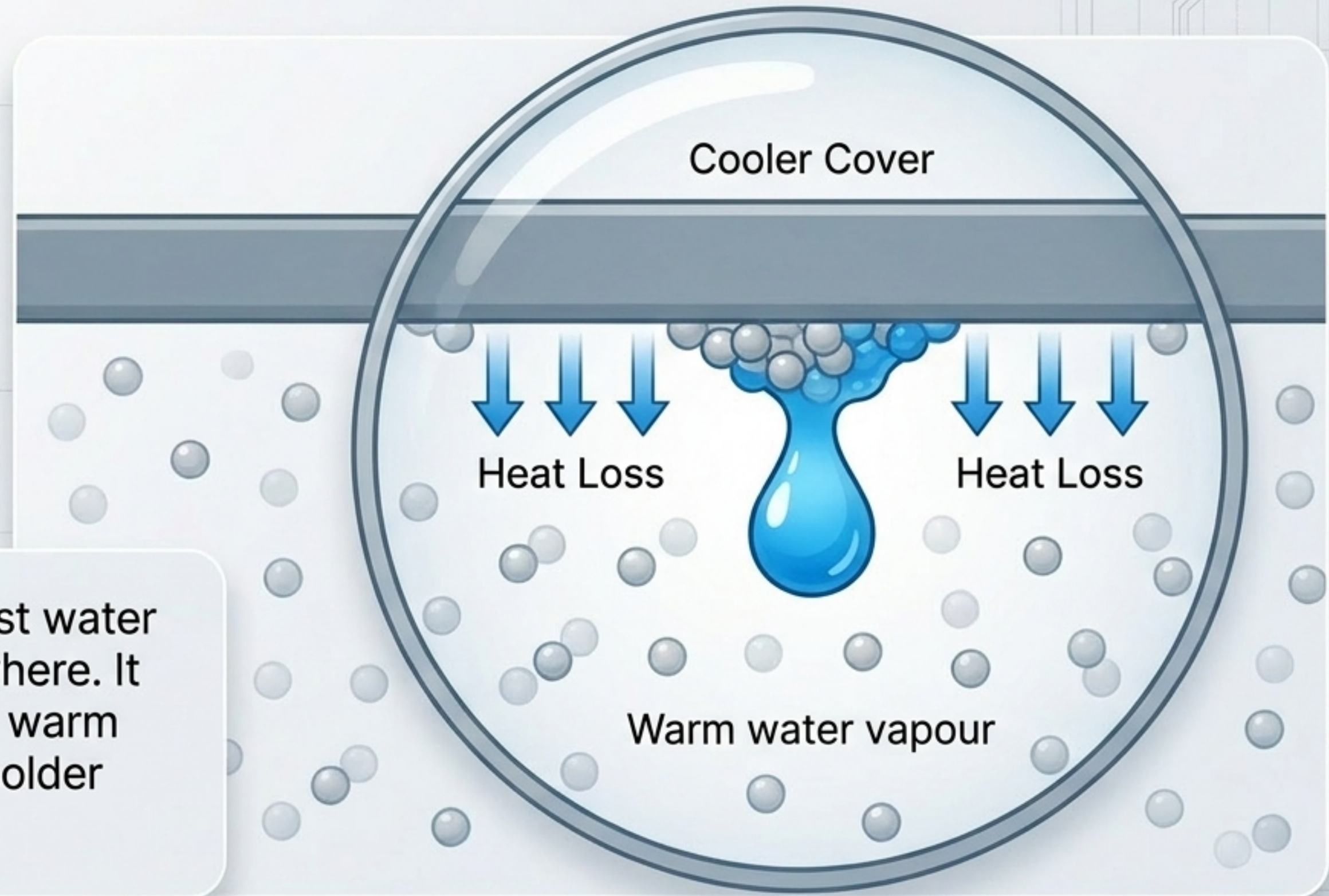
- **Change:** Liquid to Gas.
- **Location:** Occurs throughout the entire liquid.
- **Temperature:** Occurs only at a fixed boiling point.

The Condensation Mechanism



The higher the temperature difference between the warm water and the lid, the faster the first droplet falls.

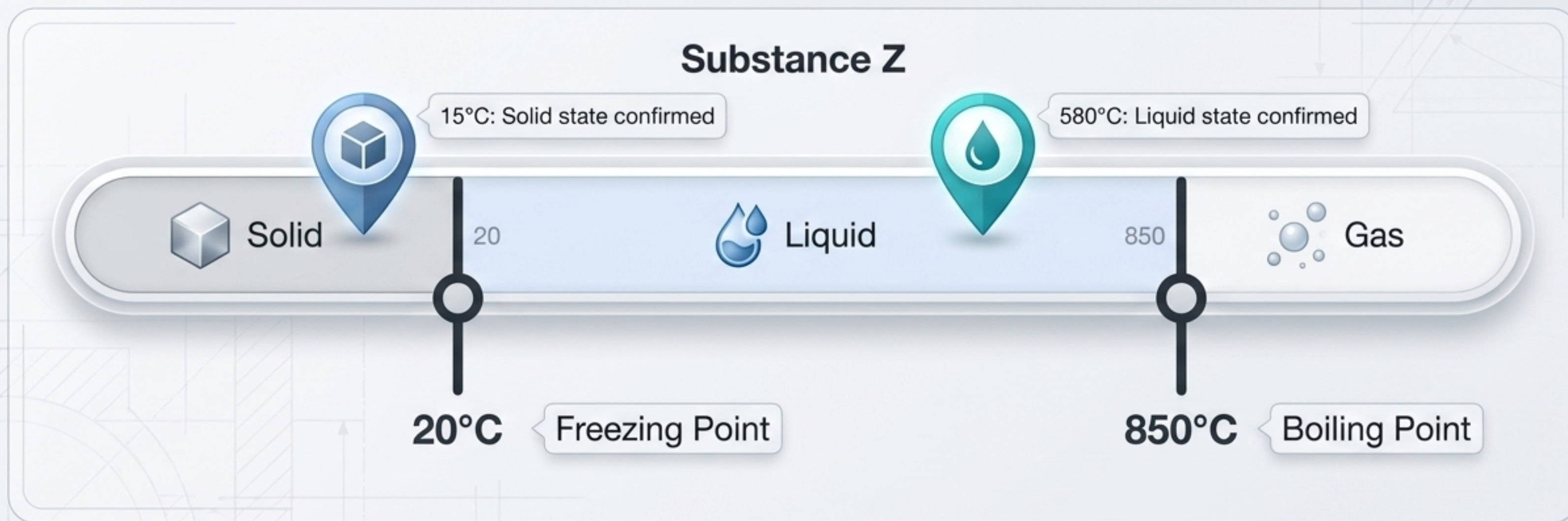
Zooming In: The Cooler Surface



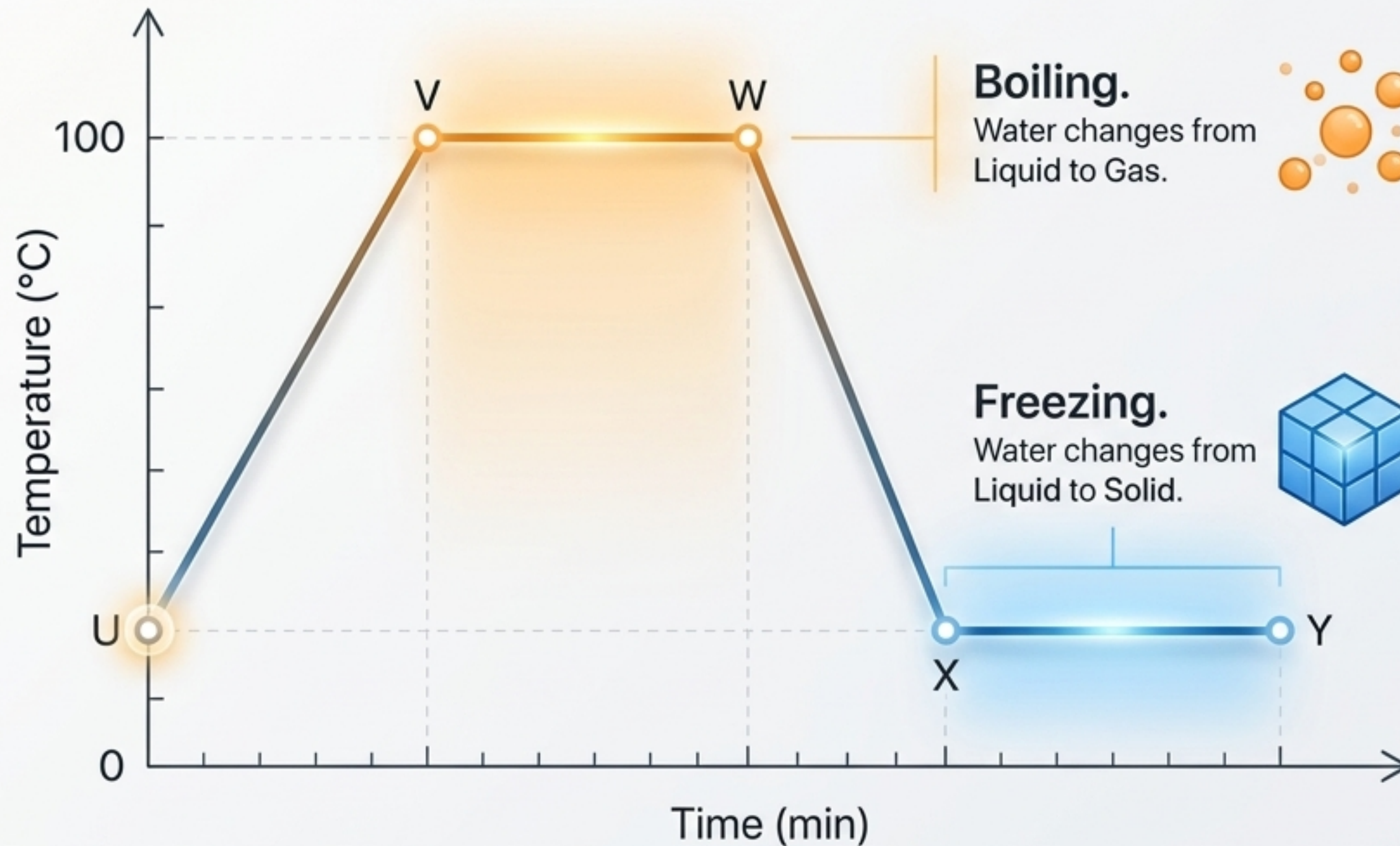
Condensation isn't just water appearing out of nowhere. It is the direct result of warm gas losing heat to a colder physical boundary.

The Fingerprint of Matter

Every substance has a specific freezing and boiling point. By knowing these two numbers, you can predict its exact state at any temperature.



The Flatline Phenomenon



The Golden Rule

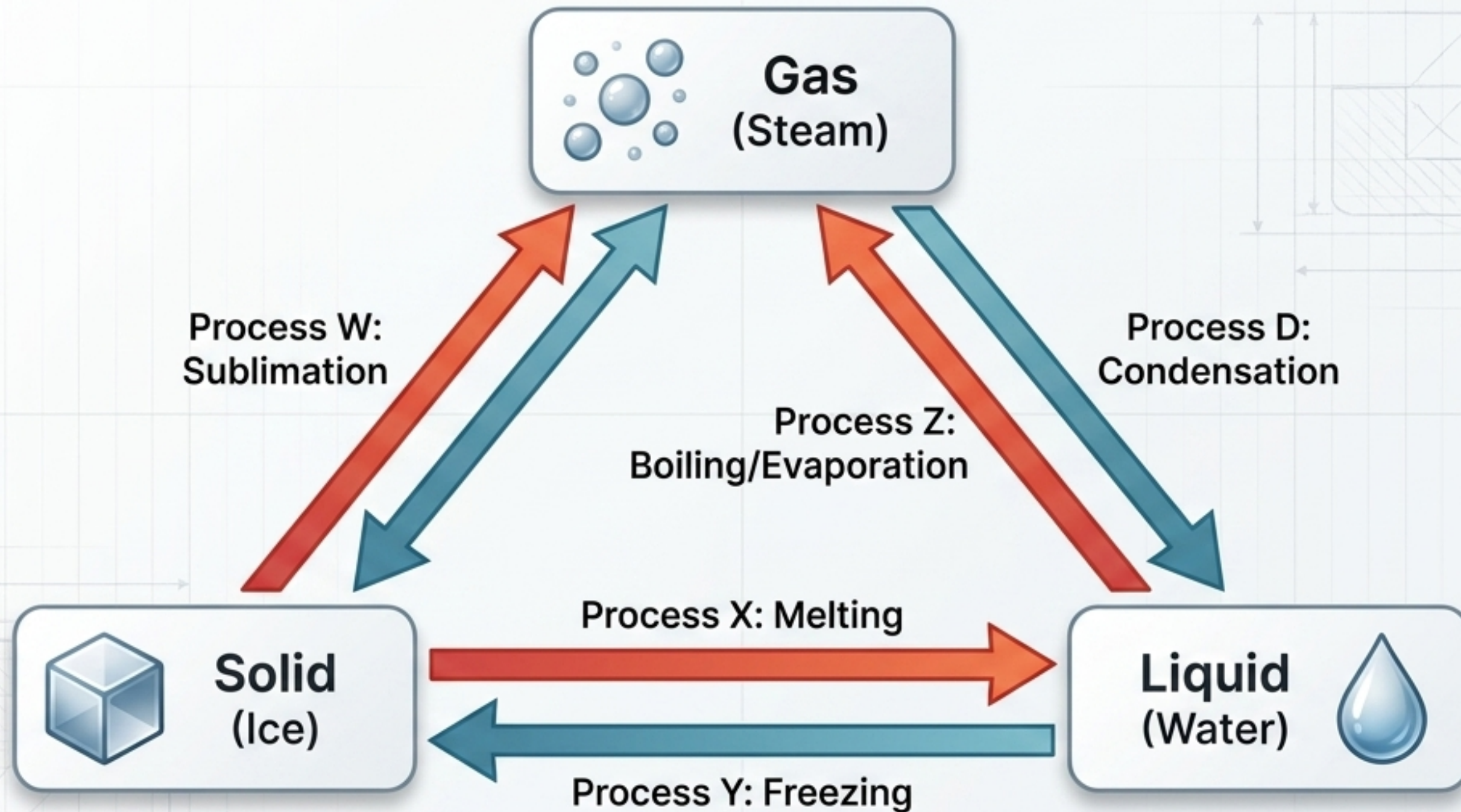
Temperature remains completely constant during a change of state.

The heat energy is used to break or form bonds, not to change the temperature.

Two consenotations of the bordinary stepson.

-  Heat Loss/ Solid
-  Heat Loss/ Liquid
-  Heat Gain/ Gas

The Master Engine of Phase Changes



Every change in the state of matter is driven by a single rule:
the exchange of heat with the surrounding environment.